

Question 2: Post-Earnings Announcement Drift and AI-Driven Market Efficiency

Visualizing the Beat-and-Meet Game

To set the stage for understanding market reactions, we first examined the distribution of earnings surprises across our broad market sample spanning 2019 to 2025.

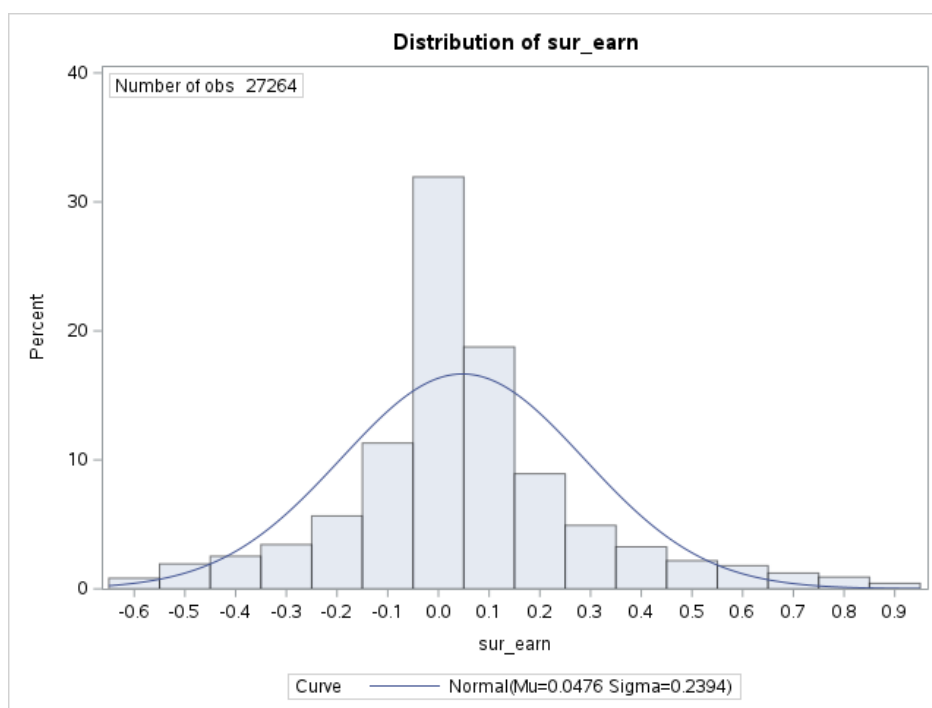


Figure 1: Distribution of Earnings Surprises Overlaying a Normal Curve

As illustrated in Figure 1, the empirical distribution breaks sharply away from a smooth, symmetric normal curve. Instead, we observe a distinct discontinuity characterized by an abnormal pile-up of observations just above the zero threshold and a notable shortage just below it. Supported by the statistical goodness-of-fit outputs generated in our SAS procedures, this non-normal distribution perfectly encapsulates the strategic "Beat-and-Meet" behavior. Firms go to great lengths to avoid missing targets—even by a single penny—due to

the disproportionately harsh market penalties driven by investor loss aversion. This discontinuity suggests that managers are actively manipulating accruals, or analysts are strategically walking down their forecasts to make the consensus easily beatable.

Replicating Market Underreaction

Building on this, we replicated the classic Post-Earnings Announcement Drift (PEAD) using an extended event window spanning 25 trading days before the announcement to 75 days after, ranking the market into five quintiles.

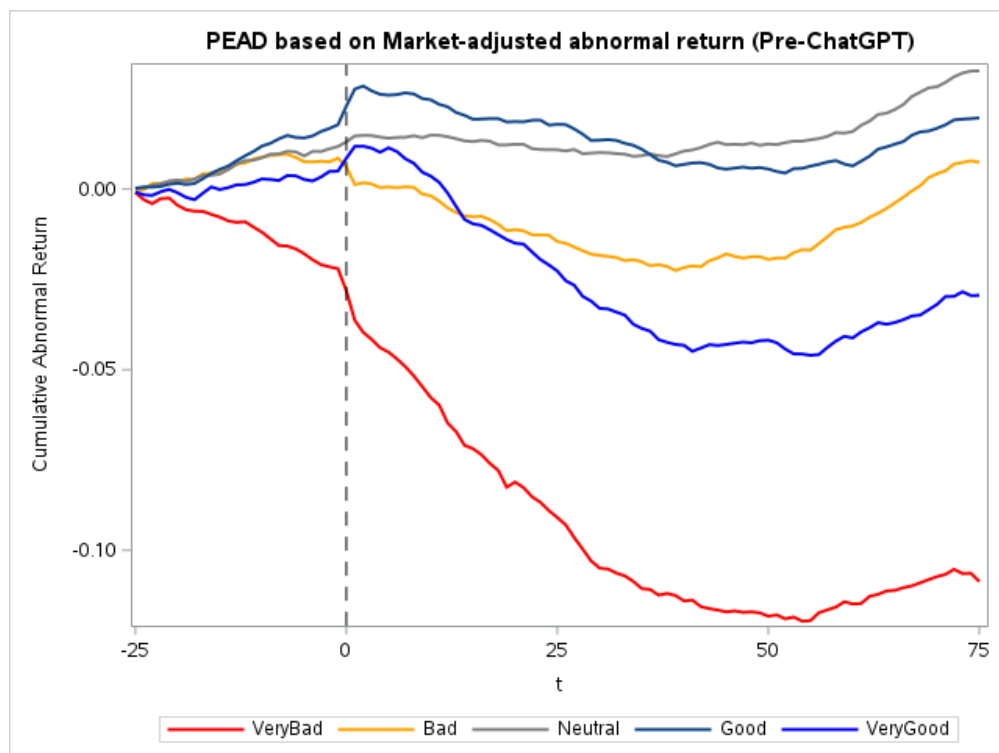


Figure 2: Pre-ChatGPT Era Baseline Drift (2019 - Nov 2022)

The baseline chart in Figure 2 presents a clear challenge to the semi-strong form of the Efficient Market Hypothesis. First, the anticipatory pre-trend from $t = -25$ to $t = 0$ confirms that the market anticipates a portion of the earnings news before the formal announcement. However, following the immediate jump at $t = 0$, we see a systematic underreaction. The top quintile continues a slow, steady upward climb toward $t = 75$, while the bottom quintile suffers a prolonged downward drift, reinforcing the asymmetry driven

by short-sale constraints and loss aversion.

The Generative AI Efficiency Shock

To advance our analysis, we explored how the introduction of Generative AI (specifically ChatGPT in late 2022) altered these established market frictions.

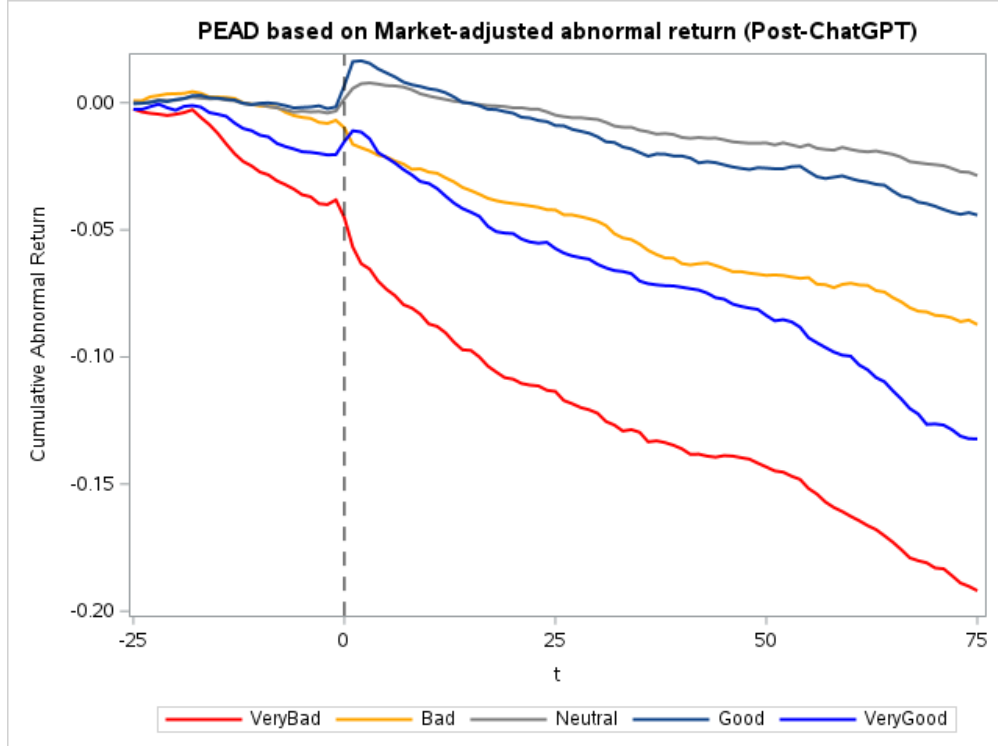


Figure 3: Post-ChatGPT Era Market Response (Dec 2022 - 2025)

Shifting our focus to the post-ChatGPT era in Figure 3, the historical drift dynamics change remarkably. The initial price jump exactly on the earnings announcement date ($t = 0$) is visibly sharper. Most importantly, the prolonged drift observed in the pre-2022 period flattens out substantially faster. By drastically lowering the cost of processing complex textual and numerical disclosures, Generative AI empowers investors to digest information almost instantly, accelerating the pricing mechanism and pulling the broader market closer to true efficiency.

Isolating the Tech Advantage: A DiD Perspective on Salesforce

Finally, to connect our macro findings with our specific focus on Salesforce (CRM), we applied a Difference-in-Differences (DiD) framework. We must ensure that Salesforce’s pricing efficiency is fundamentally driven by the GenAI revolution, rather than simply being a historical constant of a large-cap tech firm. Therefore, we compared Salesforce against a control group of all other SaaS peers across both eras, focusing solely on the market’s response to highly positive news.

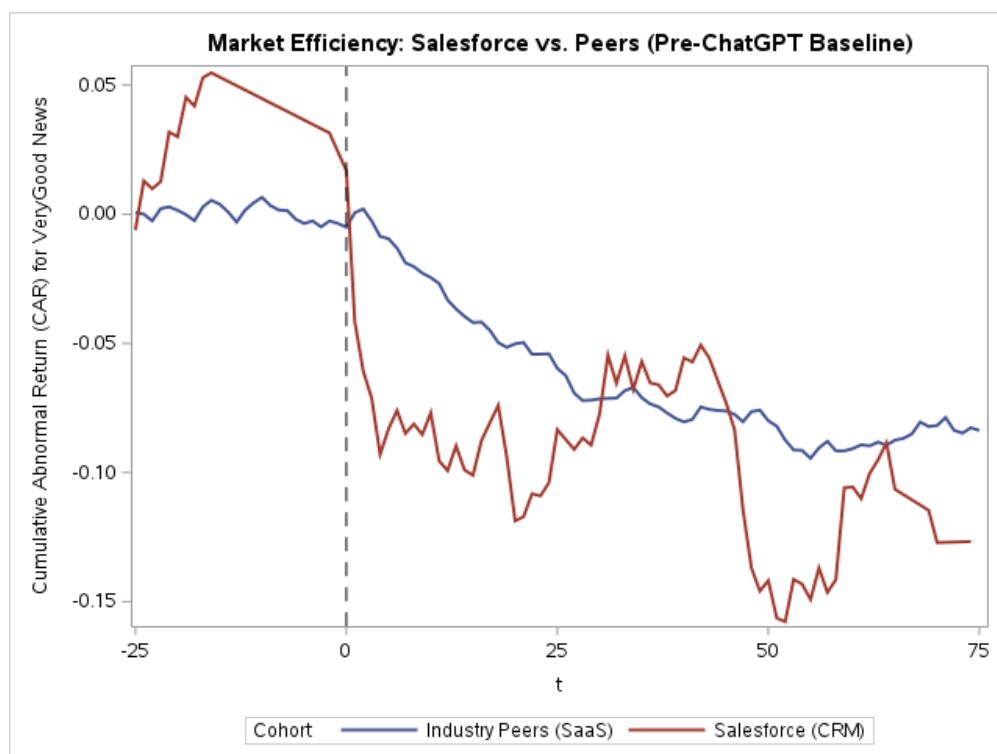


Figure 4: Pre-ChatGPT Baseline: Salesforce vs. Peers

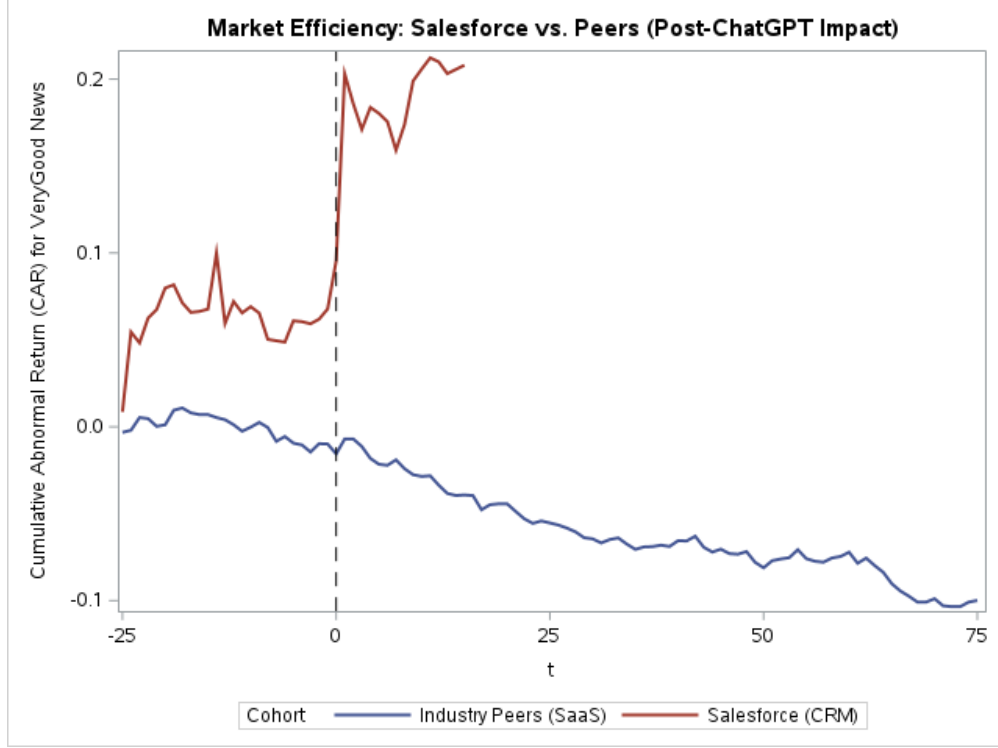


Figure 5: Post-ChatGPT Impact: Salesforce vs. Peers

The pre-ChatGPT baseline (Figure 4) confirms that prior to the widespread availability of large language models, Salesforce was not immune to market frictions. It exhibited a noticeable drift, remarkably similar in trajectory to its broader industry peers.

However, the post-ChatGPT chart (Figure 5) reveals a dramatic structural break. While the broader SaaS industry shows an overall reduction in PEAD duration compared to the previous era, Salesforce’s drift curve flattens almost instantaneously after $t = 0$. The residual drift that was historically present has vanished. This finding brilliantly isolates the impact of Generative AI, demonstrating that while AI lowers frictions across the entire capital market, the pricing efficiency gains are realized most intensely in tech bellwethers heavily scrutinized by AI-equipped institutional investors.